

What One Orbit Produces

Three Consequences of 4π Closure:
The Electromagnetic Coupling, the Magnetic Moment,
and Pauli Exclusion

SFT Combined Research Note

Shaun Fosmark
Independent researcher

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The same geometric structure that was derived for particle mass independently produces the electromagnetic coupling framework, the tree-level magnetic moment, and the Pauli exclusion principle. Not by design. By consequence.

Status. This note combines three separate investigations (Alpha-137, Magnetic Moment, MWO Pauli Chain) under their common geometric origin: the 4π spinor closure of the locally real SFT particle. Each result is conditional. Each follows from structures derived for different purposes. Their convergence on one geometric fact is the central observation.

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1 The 4π closure

In SFT, a massive particle is energy at c caught in a closed orbit. The orbit requires 4π of spatial angle to restore the energy's internal orientation — not 2π like a classical orbit. A 2π rotation returns the energy to the same position but with FLIPPED orientation. Only after 4π does both position and orientation return to the starting state.

This is spin- $\frac{1}{2}$: the frame rotates at half the spatial rate. The defining property is:

$$R(2\pi) = -1, \quad R(4\pi) = +1. \quad (1)$$

The 4π closure was derived in V13.0 from the particle's orbital geometry in 3+1 spacetime, not from quantum mechanics. It entered the theory as the explanation for why particles have mass: the closure load $4\pi_{\text{spinor}} + 2\pi_{\text{frame}} = 6\pi$ determines the compounding rate $L_{\text{eff}} = 3\pi - \frac{1}{2}\ln(25/13)$ that gives the mass hierarchy to 0.032%.

The three investigations reported here discovered — independently and in sequence — that the same 4π produces three additional consequences in the quantum domain.

2 Consequence 1: The electromagnetic coupling

SOURCE: ALPHA-137 RESEARCH NOTE, R257B–R258E.

The electromagnetic coupling $\alpha \approx 1/137$ measures how strongly charged particles interact. In SFT, charge is the circulation direction of the 4π closure — the θ_g holonomy, independent of mass cadence ϕ_C . The coupling α should be the fraction of total settlement that carries directional (charge) information.

The investigation found that directional settlement from a charged particle must project through two full 4π solid-angle boundaries to reach the exterior:

$$\text{closure sphere} \xrightarrow{4\pi} \text{boundary} \xrightarrow{4\pi} \text{exterior}. \quad (2)$$

Each projection dilutes the directional signal by $1/(4\pi)$. Two projections give a base coupling:

$$\alpha_{\text{base}} = \frac{1}{(4\pi)^2} = \frac{1}{16\pi^2} \approx \frac{1}{158}. \quad (3)$$

A scale-dependent participation correction from the 3:2 sector structure (the same structure that gives L_{eff}) modifies the base:

$$\boxed{\alpha(E) = \frac{N_{\text{eff}}^{p(E)}}{16\pi^2}, \quad N_{\text{eff}} = \frac{25}{13}.} \quad (4)$$

At low energy, $p_{\text{low}} \approx 0.217$ gives $\alpha \approx 1/137$. At the Z mass, $p_Z \approx 0.322$ gives $\alpha \approx 1/128$. The running is monotonic with energy, reproducing the qualitative behavior of the measured electromagnetic coupling across all scales.

What 4π does here: The $(4\pi)^2$ in the denominator is two solid-angle projections through the particle's 4π settlement sphere. The electromagnetic coupling is weak ($\sim 1/137$ rather than ~ 1) because the directional signal must project through the FULL 4π geometry of the closure. The same 4π that makes the particle massive also makes electromagnetism weak.

3 Consequence 2: The magnetic moment

SOURCE: MAGNETIC MOMENT RESEARCH NOTE, R276M–R276H.

The electron magnetic moment is measured as $\mu = g(q/2m)S$ with $g = 2.00231930436\dots$. The tree-level value $g = 2$ is not a parameter in SFT. It is a consequence of 4π closure.

3.1 What fails

A naive classical current loop at the SFT spin radius $r_{\text{spin}} = \hbar/(2mc)$ gives:

$$\mu_{\text{loop}} = \frac{q\hbar}{4m} = \frac{\mu_B}{2}. \quad (5)$$

Half the correct value. The naive model fails by exactly the factor that 4π closure must supply.

3.2 What works

The particle writes at the leading edge of its own history. The future is unwritten. The current write has access only to local settlement data. Therefore the next-write operator must be first-order in the complete current support state — second-order data would require information from unwritten territory.

A first-order operator whose square gives the one-speed mass-shell $E^2 = p^2c^2 + m^2c^4$ must satisfy:

$$\{\Gamma^\mu, \Gamma^\nu\} = 2\eta^{\mu\nu}I. \quad (6)$$

This is the Clifford algebra — the unique algebraic structure whose square gives the Minkowski metric. Combined with 4π spinor closure in three dimensions, the generators are the Pauli matrices σ_i .

Coupling the θ_g charge holonomy through the covariant derivative $D_\mu = \partial_\mu - i(q/\hbar)A_\mu$ produces the curvature commutator:

$$[D_\mu, D_\nu] = -i\frac{q}{\hbar}F_{\mu\nu}. \quad (7)$$

The squared first-order operator then generates the Pauli spin-field coupling:

$$(\Gamma \cdot \Pi)^2 = \Pi^2 - \frac{q\hbar}{2}\sigma^{\mu\nu}F_{\mu\nu}, \quad (8)$$

giving magnetic interaction $H_{\text{spin}} = -(q/m)\mathbf{S} \cdot \mathbf{B}$ and therefore:

$$\boxed{g_{\text{tree}} = 2.} \quad (9)$$

What 4π does here: The 4π closure selects the Clifford/Pauli algebra, which forces the spin-field coupling to have the SAME strength as the charge-field coupling. The electric response and the magnetic spin response are two projections of one θ_g -coupled 4π closure. Same source, same coupling. $1 + 1 = 2$.

3.3 The deeper connection

Uncertainty, first-order dynamics, and $g = 2$ are three views of one constraint: the particle writes at the leading edge. Leading-edge access means first-order. First-order on a 4π spinor gives Clifford. Clifford with θ_g coupling gives the Pauli term. The leading edge that creates uncertainty also creates the magnetic moment.

3.4 The anomalous moment

The measured $g = 2(1 + a_e)$ with $a_e \approx \alpha/(2\pi)$ is naturally interpreted as a self-settlement correction: the electron's own past θ_g writes modifying the current write geometry. The correction is proportional to α (the directional coupling from Sec. 2). The anomalous moment connects the α program to the magnetic moment program through the same 4π closure. It is open.

4 Consequence 3: Pauli exclusion

SOURCE: MWO INTERNAL NOTE 03, MWO-04 THROUGH MWO-23.

Two identical electrons cannot occupy the same complete quantum state. This is the foundation of atomic shell structure, the periodic table, and all of chemistry. In SFT, it follows from 4π closure plus no-overwrite.

4.1 Duplicate rejection

No-overwrite means a completed write cannot be overwritten. Applied to quantum ledger addresses: the exact same complete state cannot be written twice. If W_a is the write operator for state a :

$$W_a^2 = 0. \quad (10)$$

This is necessary but not sufficient. It blocks same-state double occupation without explaining why DISTINCT-state writes anticommute.

4.2 Oriented writes

A 4π write is not blank occupancy. It carries orientation from the spinor closure. Two oriented writes placed on the settlement ledger form an oriented boundary — a patch with a definite “which side is up.” Reversing the order of two oriented writes flips the patch. Flipping an oriented patch changes the sign:

$$W_a W_b = -W_b W_a. \quad (11)$$

The commuting nilpotent countermodel ($W_a^2 = 0$ with $W_a W_b = +W_b W_a$) is killed because letting oriented writes commute would erase the orientation that defines them as 4π closures.

4.3 The N-particle ledger

The two-write rule extends to N identical 4π supports through the exterior product:

$$\Psi_N = e_1 \wedge e_2 \wedge \cdots \wedge e_N. \quad (12)$$

Any pair swap flips sign: $P_{ij}\Psi_N = -\Psi_N$. Any duplicate collapses: $e_1 \wedge e_2 \wedge e_1 = 0$.

Identical 4π writes: $W_a W_b = -W_b W_a, \quad W_a^2 = 0.$

(13)

What 4π does here: The 4π closure gives the ORIENTATION that makes writes anticommute rather than commute. A 2π (bosonic) closure has no orientation mismatch under exchange — its writes commute and share states freely. The 4π closure's half-integer twist forces antisymmetry forces exclusion forces shell structure forces the periodic table.

4.4 Entanglement as a byproduct

The investigation also produced a native SFT entanglement mechanism: at pair production, two photons meet and wrap around each other AFTER contact (causality forbids pre-contact reaction). The post-contact braid is a permanent feature of the written worldline history. Entanglement is shared formation history, not a spooky later connection. No-overwrite makes the braid permanent.

5 The pattern

One geometric fact: the particle's energy completes a 4π spinor closure in 3+1 dimensions.

Three consequences:

Domain	What 4π produces	Result
EM coupling	Two-stage $(4\pi)^2$ projection	$\alpha \approx 1/(16\pi^2 \times \text{correction})$
Magnetic moment	Clifford algebra from 4π	$g_{\text{tree}} = 2$
Statistics	Oriented ledger from 4π	$W_a W_b = -W_b W_a$, exclusion

None of these consequences were anticipated when 4π closure was derived in V13.0 for particle mass. The mass hierarchy derivation required 4π as the spatial curvature crossed per Compton period. The electromagnetic coupling, magnetic moment, and Pauli exclusion were discovered later — independently, in sequence, each time by following the mathematics of 4π closure into a domain where it had not previously been applied.

The theory was not engineered to produce these results. It was engineered to produce the mass hierarchy. Everything else was a consequence the geometry already contained.

6 Claim boundary

Earned:

The α projection kernel $1/(16\pi^2)$ is the robust geometric base, with monotonic running across twelve orders of magnitude in energy.

The tree-level $g = 2$ follows conditionally from leading-edge first-order dynamics on a 4π θ_g -coupled closure.

Local and N -particle Pauli exclusion follows from oriented 4π writes anticommuting under no-overwrite.

Pair-production braid as the native SFT entanglement mechanism.

Not earned:

The exact α running function $p(E)$.

The anomalous magnetic moment $a_e = (g-2)/2$.

The full QFT spin-statistics theorem.

The formal first-order closure operator Γ^μ .

The microscopic settlement operator that unifies all three consequences.

The common open target:

All three results are conditional on the same missing engine: the microscopic settlement operator. The operator that determines how 4π closure propagates, self-interacts, and compounds. Its

derivation would simultaneously close the α running, the magnetic moment formalization, and the spin-statistics proof.

Three consequences. One open core. The geometry is identified. The operator is the mountain.

The sentence

4π spinor closure, derived for particle mass, independently produces the electromagnetic coupling framework, the tree-level electron magnetic moment, and the Pauli exclusion principle — three pillars of quantum mechanics from one geometric fact about how energy orbits in 3+1 dimensions.

Acknowledgments

The Alpha-137 investigation (R257B–R258E) was developed with ChatGPT/5.5 (OpenAI) and Claude/Opus (Anthropic). The magnetic moment investigation (R276M–R276H), including the leading-edge unification of uncertainty, first-order dynamics, and $g = 2$, was developed in conversation between Shaun Fosmark and Claude/Opus, with route verification by ChatGPT/5.5. The MWO Pauli chain (MWO-04 through MWO-23), including the pair-production braid, the history-tag firewall, and the oriented exterior ledger, was developed by all three collaborators through twenty-three adversarial gates.

References

- [1] S. Fosmark, *Solvency Field Theory V13.1: An Information-Theoretic Foundation for Cosmology and Gravity*, Zenodo (2026). <https://zenodo.org/records/20822219>